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DEPARTAMENTO DE ENGENHARIA ELETROTÉCNICA E DE COMPUTADORES

Master's Degree in Electrical and Computer Engineering

DIGITAL SYSTEM PROJECT

PROJECT TITLE

Design and Implementation of a Pong Game on FPGA Using VHDL

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Introduction and Objective

To design a hardware-based implementation of the classic Pong arcade game using VHDL, with graphical output via VGA, user input through PS/2 keyboard, and deployed on the Altera DE2 development board with a Cyclone II EP2C35F672C6 FPGA.

Tools and Platform

- Hardware: Altera DE2 Board (Cyclone II EP2C35F672C6)
- Software: Quartus II (v13.0 or compatible with DE2)
- Language: VHDL (structural and behavioral)

System Architecture

The project is composed of the following VHDL modules:

1. VGA Sync Generator: Generates VGA timing signals for 640×480
Outputs:
hsync, vsync: synchronization signals
video_on: signal to validate visible area
pixel_x, pixel_**y**: current pixel coordinates

2. PS/2 Keyboard Interface: This reads scancodes from a standard PS/2 keyboard. It also converts scancodes to movement commands (e.g., up, down). Debounces and handles key-press logic.

3. Game Logic Controller
- Handles game rules: ball movement, paddle movement, collisions.
- Inputs from keyboard interface.
- Maintains state machines for the ball and paddles.
- Manages score logic.

4. Object Rendering Module
- Encodes paddle and ball positions into RGB pixel data.
- Outputs:
- rgb: 3-bit or 8-bit signal for VGA DAC
- pixel_on: boolean signal to indicate active drawing

5. Top-Level Module
- Instantiates all submodules.
- Connects inputs (clock, keyboard, VGA output) and I/O pins.

PS/2 Keyboard Integration

Supported Keys:
- W/S – Move left paddle up/down
- Up/Down Arrows – Move right paddle up/down
- PS/2 state machine handles:
- Start bit
- 8-bit scancode
- Parity and stop bits

A finite state machine decodes scancodes and maps them to movement logic for the paddles.

Game Logic Details

Ball Movement:

- Ball moves diagonally at constant speed.
- Reverses direction on:
- Paddle collisions
- Top/bottom wall collisions
- Miss results in a score update and ball reset.

Paddle Control:

- Limited to vertical movement.
- Paddle positions updated based on keyboard input.

Scoring:

- Optional 7-segment score display
- Internal score counter resets upon player miss.

Testing and Verification

Procedure:

- Compile in Quartus
- Load .sof onto DE2 board via JTAG
- Connect VGA monitor and PS/2 keyboard
- Confirm:
- Stable VGA display at 640×480
- Paddle movement via keys
- Ball movement and collision detection

Results

- Stable VGA output without flickering
- Real-time paddle control via PS/2 keyboard
- Functional ball bouncing and collision with paddle/wall
- Reset and score update logic working correctly

Conclusion

This project demonstrates a successful implementation of a hardware arcade game entirely in VHDL, with direct I/O integration and real-time graphics using the VGA protocol. It validates practical FPGA design skills including FSMs, peripheral interfacing, timing control, and modular hardware development.REP